

● HARD

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

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# Geometry and Trigonometry

# 02

10 answers · Rationale-rich · Teach from doc

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

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Answer not provided in source data — see rationale below.

The correct answer is  $\frac{1}{6}$ . The circumference,  $C$ , of a circle is  $C = 2\pi r$ , where  $r$  is the length of the radius of the circle. For the given circle with a radius of 1, the circumference is  $C = 2(\pi)(1)$ , or  $C = 2\pi$ . To find what fraction of the circumference the length of arc  $\overline{AB}$  is, divide the length of the arc by the circumference, which gives  $\frac{\pi}{3} \div 2\pi$ . This division can be represented by  $\frac{\pi}{3} \cdot \frac{1}{2\pi} = \frac{1}{6}$ . Note that  $1/6$ ,  $.1666$ ,  $.1667$ ,  $0.166$ , and  $0.167$  are examples of ways to enter a correct answer.

Q2

24

The correct answer is 24. The sine of an acute angle in a right triangle is equal to the ratio of the length of the side opposite the angle to the length of the hypotenuse. In the triangle shown, the sine of angle B, or  $\sin(B)$ , is equal to the ratio of the length of side  $\overline{AC}$  to the length of side  $\overline{AB}$ . It's given that the length of side  $\overline{AB}$  is 26 and that  $\sin(B) = \frac{5}{13}$ . Therefore,  $\frac{5}{13} = \frac{AC}{26}$ . Multiplying both sides of this equation by 26 yields  $AC = 10$ .

By the Pythagorean Theorem, the relationship between the lengths of the sides of triangle ABC is as follows:  $26^2 = 10^2 + BC^2$ , or  $676 = 100 + BC^2$ . Subtracting 100 from both sides of  $676 = 100 + BC^2$  yields  $576 = BC^2$ . Taking the square root of both sides of  $576 = BC^2$  yields  $24 = BC$ .

The correct answer is 123. The triangle angle sum theorem states that the sum of the measures of the interior angles of a triangle is 180 degrees. It's given that the measure of  $\angle SQX$  is  $48^\circ$  and the measure of  $\angle SXQ$  is  $86^\circ$ . Since points  $S$ ,  $Q$ , and  $X$  form a triangle, it follows from the triangle angle sum theorem that the measure, in degrees, of  $\angle QSX$  is  $180 - 48 - 86$ , or 46. It's also given that the measure of  $\angle SWU$  is  $85^\circ$ . Since  $\angle SWU$  and  $\angle SWR$  are supplementary angles, the sum of their measures is 180 degrees. It follows that the measure, in degrees, of  $\angle SWR$  is  $180 - 85$ , or 95. Since points  $R$ ,  $S$ , and  $W$  form a triangle, and  $\angle RSW$  is the same angle as  $\angle QSX$ , it follows from the triangle angle sum theorem that the measure, in degrees, of  $\angle WRS$  is  $180 - 46 - 95$ , or 39. It's given that the measure of  $\angle VTU$  is  $162^\circ$ . Since  $\angle VTU$  and  $\angle STU$  are supplementary angles, the sum of their measures is 180 degrees. It follows that the measure, in degrees, of  $\angle STU$  is  $180 - 162$ , or 18. Since points  $R$ ,  $T$ , and  $U$  form a triangle, and  $\angle URT$  is the same angle as  $\angle WRS$ , it follows from the triangle angle sum theorem that the measure, in degrees, of  $\angle TUR$  is  $180 - 39 - 18$ , or 123.

**Q4****C****C. 12**

Choice C is correct. It's given that rectangle  $ABCD$  is similar to rectangle  $EFGH$ . Therefore, if the length of each side of rectangle  $ABCD$  is  $k$  times the length of the corresponding side of rectangle  $EFGH$ , then the area of rectangle  $ABCD$  is  $k^2$  times the area of rectangle  $EFGH$ . It's given that the area of rectangle  $ABCD$  is 648 square inches and the area of rectangle  $EFGH$  is 72 square inches. It follows that  $k^2 = \frac{648}{72}$ , or  $k^2 = 9$ . Taking the square root of each side of this equation yields  $k = \sqrt{9}$ , or  $k = 3$ . It follows that the length of each side of rectangle  $ABCD$  is 3 times the length of the corresponding side of rectangle  $EFGH$ . It's given that the length of the longest side of rectangle  $ABCD$  is 36 inches. Therefore, 36 inches is 3 times the length of the longest side of rectangle  $EFGH$ , and the longest side of rectangle  $EFGH$  is equal to  $\frac{36}{3}$ , or 12, inches.

Choice A is incorrect. This is the length, in inches, of the longest side of a rectangle with side lengths that are  $\frac{1}{9}$  the corresponding side lengths of rectangle  $ABCD$ , rather than a rectangle with an area that is  $\frac{1}{9}$  the area of rectangle  $ABCD$ .

Choice B is incorrect. This is the factor by which the area of rectangle  $ABCD$  is larger than the area of rectangle  $EFGH$ , not the length, in inches, of the longest side of rectangle  $EFGH$ .

Choice D is incorrect. This is the length, in inches, of the longest side of rectangle  $ABCD$ , not rectangle  $EFGH$ .

**Q5****11**

The correct answer is 11. A circle with equation  $(x - a)^2 + (y - b)^2 = r^2$ , where  $a$ ,  $b$ , and  $r$  are constants, has center  $(a, b)$  and radius  $r$ . Therefore, the radius of the given circle is  $\sqrt{121}$ , or 11.

**Q6****D**

D. 250

Choice D is correct. Corresponding angles in similar triangles have equal measures. It's given that triangle  $ABC$  is similar to triangle  $DEF$ , where  $A$  corresponds to  $D$ , so the measure of angle  $A$  is equal to the measure of angle  $D$ . Therefore, if  $\tan A = \sqrt{3}$ , then  $\tan D = \sqrt{3}$ . It's given that angles  $C$  and  $F$  are right angles, so triangles  $ABC$  and  $DEF$  are right triangles. The adjacent side of an acute angle in a right triangle is the side closest to the angle that is not the hypotenuse. It follows that the adjacent side of angle  $D$  is side  $DF$ . The opposite side of an acute angle in a right triangle is the side across from the acute angle. It follows that the opposite side of angle  $D$  is side  $EF$ . The tangent of an acute angle in a right triangle is the ratio of the length of the opposite side to the length of the adjacent side. Therefore,  $\tan D = \frac{EF}{DF}$ . If  $DF = 125$ , the length of side  $EF$  can be found by substituting  $\sqrt{3}$  for  $\tan D$  and 125 for  $DF$  in the equation  $\tan D = \frac{EF}{DF}$ , which yields  $\sqrt{3} = \frac{EF}{125}$ . Multiplying both sides of this equation by 125 yields  $125\sqrt{3} = EF$ . Since the length of side  $EF$  is  $\sqrt{3}$  times the length of side  $DF$ , it follows that triangle  $DEF$  is a special right triangle with angle measures  $30^\circ$ ,  $60^\circ$ , and  $90^\circ$ . Therefore, the length of the hypotenuse,  $DE$ , is 2 times the length of side  $DF$ , or  $DE = 2DF$ . Substituting 125 for  $DF$  in this equation yields  $DE = 2(125)$ , or  $DE = 250$ . Thus, if  $\tan A = \sqrt{3}$  and  $DF = 125$ , the length of  $DE$  is 250.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the length of  $EF$ , not  $DE$ .

**Q7****153**

The correct answer is 153. Since it's given that  $\overline{PQ}$  is parallel to  $\overline{XY}$  and angle  $Y$  is a right angle, angle  $ZQP$  is also a right angle. Angle  $ZPQ$  is complementary to angle  $XZY$ , which means its measure, in degrees, is  $90 - 63$ , or 27. Since angle  $XPQ$  is supplementary to angle  $ZPQ$ , its measure, in degrees, is  $180 - 27$ , or 153.

**Q8****1260**

The correct answer is 1,260. Since it's given that prisms X and Y are similar, all the linear measurements of prism Y are  $k$  times the respective linear measurements of prism X, where  $k$  is a positive constant. Therefore, the surface area of prism Y is  $k^2$  times the surface area of prism X and the volume of prism Y is  $k^3$  times the volume of prism X. It's given that the surface area of prism Y is  $1,450 \text{ cm}^2$ , and the surface area of prism X is  $58 \text{ cm}^2$ , which implies that  $1,450 = 58k^2$ . Dividing both sides of this equation by 58 yields  $\frac{1,450}{58} = k^2$ , or  $k^2 = 25$ . Since  $k$  is a positive constant,  $k = 5$ . It's given that the volume of prism Y is  $1,250 \text{ cm}^3$ . Therefore, the volume of prism X is equal to  $\frac{1,250}{k^3} \text{ cm}^3$ , which is equivalent to  $\frac{1,250}{5^3} \text{ cm}^3$ , or  $10 \text{ cm}^3$ . Thus, the sum of the volumes, in  $\text{cm}^3$ , of the two prisms is  $1,250 + 10$ , or 1,260.

**Q9****B****B. -14**

Choice B is correct. An equation of the form  $x - h^2 + y - k^2 = r^2$ , where  $h$ ,  $k$ , and  $r$  are constants, represents a circle in the  $xy$ -plane with center  $h, k$  and radius  $r$ . Therefore, the circle represented by the given equation has center  $-4, 19$  and radius 11. Since the center of the circle has an  $x$ -coordinate of  $-4$  and the radius of the circle is 11, the least possible  $x$ -coordinate for any point on the circle is  $-4 - 11$ , or  $-15$ . Similarly, the greatest possible  $x$ -coordinate for any point on the circle is  $-4 + 11$ , or 7. Therefore, if the point  $a, b$  lies on the circle, it must be true that  $-15 \leq a \leq 7$ . Of the given choices, only  $-14$  satisfies this inequality.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

## Q10

12

The correct answer is 12. It is given that the triangle has angle measures of  $30^\circ$ ,  $60^\circ$ , and  $90^\circ$ , and so the triangle is a special right triangle. The side measures of this type of special triangle are in the ratio  $2:1:\sqrt{3}$ . If  $x$  is the measure of the shortest leg, then the measure of the other leg is  $\sqrt{3}x$  and the measure of the hypotenuse is  $2x$ . The perimeter of the triangle is given to be  $18+6\sqrt{3}$ , and so the equation for the perimeter can be written as  $2x+x+\sqrt{3}x=18+6\sqrt{3}$ . Combining like terms and factoring out a common factor of  $x$  on the left-hand side of the equation gives  $(3+\sqrt{3})x=18+6\sqrt{3}$ . Rewriting the right-hand side of the equation by factoring out 6 gives  $(3+\sqrt{3})x=6(3+\sqrt{3})$ . Dividing both sides of the equation by the common factor  $(3+\sqrt{3})$  gives  $x=6$ . The longest side of the right triangle, the hypotenuse, has a length of  $2x$ , or  $2(6)$ , which is 12.

### Notes

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

**Answers recovered from rationale** for Q1 — the source CB data omitted a structured key for these items, so the answer slot shows —. The worked solution in the rationale below each entry contains the correct value.

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